California State University, Long Beach

Department of Mechanical and Aerospace Engineering

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Lab Report

By

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Torsion Test of Metals and Polymers

Course Number: MAE 374

Section Number: 3

Class Number: 5963

Instructor: Dr. Shamim Mirza

Objective:

Develop a shear stress-strain diagram for metal samples in the elastic range and determine the experimental shear modulus of elasticity of the metal samples.

Apparatus:



Figure 1.1: Digital Caliper



Figure 1.2: Ruler



Figure 1.3: Tinus Olson Lo-Torq Torsion Testing Machine

Samples:



Figure 2: Sample #1 and #2

Procedure:

1. Turn on Lo-Torq Torsion Machine.
2. Load metal rod into machine so that it is level and straight when closing the machine.
3. Tighten the notches on the rod so that the twist of the rod almost perfectly lines up with the machine.
4. Zero out all the parameters recorded on the Torsion machine.
5. Press Step.

6. When the machine is ready, flip the switch to have it start in the forward direction.

7. Let it run until it reaches 3 degrees.

8. If the line the machine is plotting isn't linear, flip the switch to stop and restart.

9. Once the angle reaches 3 degrees, flip the switch to stop.

10. Flip to switch from forward to the reverse direction.

11. Start the machine again and have it run till it's zero degrees.

12. Before removing rod from machine, use sharpie to mark where the machine was holding the rod, and calipers to measure the diameter in 3 different spots of the rod between the marked lines.

13. Press print on the machine to receive the graphs and remove the rod.

14. As the graph gives only max exact values, divide the max of both torque and angle by the number of squares reaching the point then divide the total squares by 5. Those values can be used in order to create 5 data points for the table calculations.

Test Results:

Part I. ALUMINUM ROD CALCULATIONS

Table 1. Torsion of Aluminum Rod

n	Torque Value T	Angle θ	Shear Stress	Shear Strain	Modulus of Elasticity
	In. Lbs	Degrees	$\tau = \frac{Tc}{J}$ psi	$\gamma = \frac{c\theta}{L}$ rads	$G = \frac{\tau}{\gamma}$psi
1	101.107	0.60073	1210.871	0.02657	45566.75761
2	202.214	1.20146	2421.742	0.05315	45566.75761
3	303.321	1.80220	3632.612	0.07972	45566.75761
4	404.428	2.40293	4843.483	0.10629	45566.75761
5	505.534	3.00375	6054.342	0.13287	45565.30218
				Average of All G	45566.46653

Table 2. Constants of Aluminum Rod

Constant	Value
Radius - c	0.376
Polar Moment of Inertia - J	0.0314

Length of Object - L (in)	8.5

Demonstration of Equation used in Table 2

Polar Moment of Inertia

$$J = \frac{\pi c^4}{2}$$

Example of Equation for Aluminum Rod

$$J = \frac{\pi(0.376^4)}{2}$$

Demonstration of Equations used in Table 1

Shear Stress Equation

$$\tau = \frac{Tc}{J}$$

Example of Equation for Aluminum Value 5

$$\tau = \frac{(505.534)(0.376)}{(0.0314)} = 22.362$$

Shear Strain Equation

$$\gamma = \frac{c\theta}{L}$$

Example of Equation for Aluminum Value 5

$$\gamma = \frac{(0.376)(3.00375)}{(8.5)} = 0.133$$

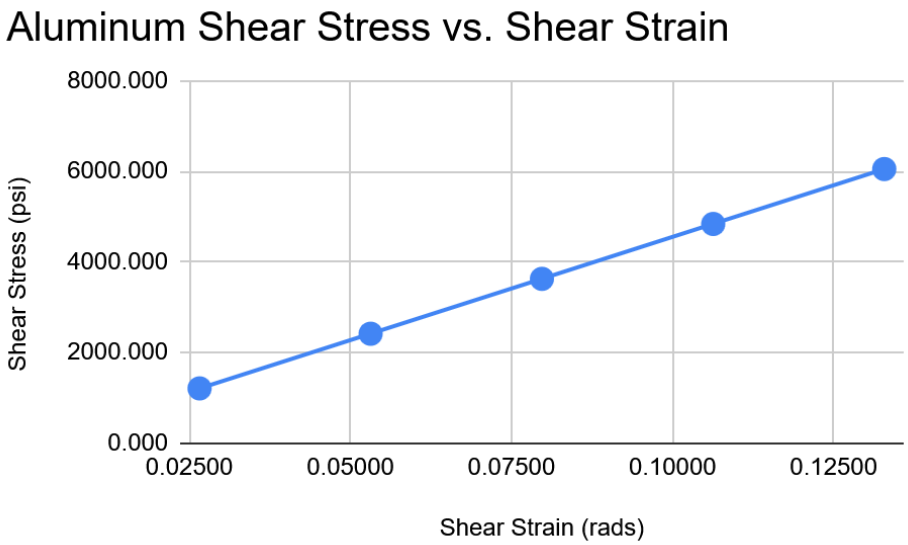
Shear Modulus of Elasticity Equation

$$G = \frac{\tau}{\gamma}$$

Example of Equation for Aluminum Value 5

$$G = \frac{(22.362)}{(0.133)} = 168.301$$

Figure 3. Shear Stress vs. Shear Strain of Aluminum



Part II. STEEL ROD CALCULATIONS

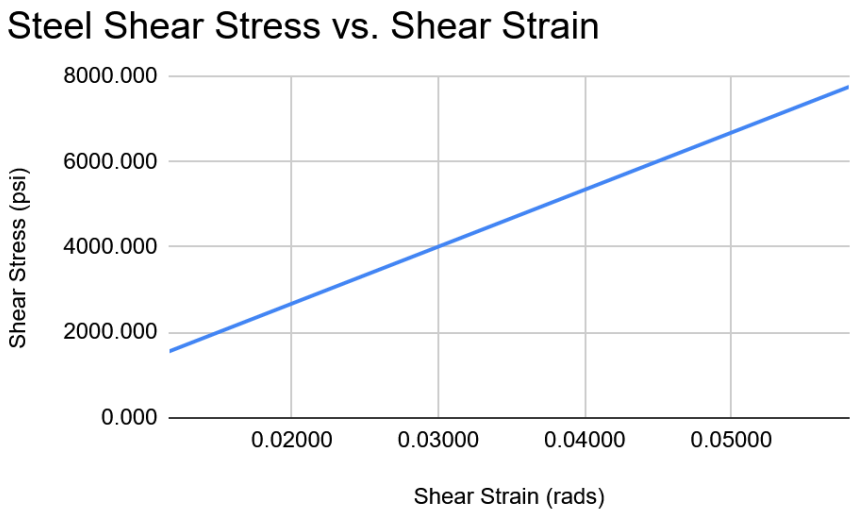
Table 3. Torsion of Steel Rod

n	Torque Value T	Angle θ	Shear Stress	Shear Strain	Modulus of Elasticity
	In. Lbs.	Degrees	$\tau = \frac{Tc}{J}$psi	$\gamma = \frac{c\theta}{L}$rads	$G = \frac{\tau}{\gamma}$psi
1	128.186	0.40275	1551.207	0.01161	133626.8558
2	256.372	0.8055	3102.413	0.02322	133626.8558
3	384.558	1.20825	4653.620	0.03483	133626.8558
4	512.744	1.611	6204.826	0.04643	133626.8558
5	640.918	2.01375	7755.888	0.05804	133624.3539
				Average of All G	133626.3554

Table 4. Constants of Steel Rod

Constant	Value
Radius - c	0.3747
Polar Moment of Inertia - J	0.0310
Length of Object - L (in)	13

Figure 4. Shear Stress vs. Shear Strain of Steel



Equations for Percent Error of Modulus of Elasticity

Percent Error Modulus of Elasticity Aluminum

Real Elasticity of Aluminum: 10 Mpsi = 10000000 psi

Percent Error Modulus of Elasticity Steel

Real Elasticity of Steel ~29 Mpsi = 29000000 psi

Results and Discussion:

The results lead to the statement that steel material is much stiffer than aluminum. This was determined solely from the comparison between the steel and aluminum shear stress (σ) and shear strain (ε) graphs. Referencing figures 3 and 4, the comparison in the steepness of the graphed values is a visual indicator of the hardness the material possesses. With steel displaying a much steeper line than aluminum. Not only is the visual confirmation of this in the graphs, but it is also emphasized by the calculation of the modulus of elasticity. The modulus of elasticity (G) tells the stiffness of a material and its resistance to deformation when a force is applied. The ratio of G for steel and aluminum results in steel having an elasticity around 3 times larger than aluminum. These results provide a clear relation between the modulus of elasticity and variations of its value across materials.

Answers to Questions:

1. Specify three assumptions that have been made in deriving the equations used in this experiment.
The material is assumed to be homogeneous and isotropic. The shaft is assumed to remain within the elastic range and obey Hooke's law. It is also assumed that plane cross-sections remain plain and circular after twisting.

2. Name three limitations of the torsion formula.
The torsion formula is applicable to circular shafts, assumes elastic material behavior, and assumes the shaft has a uniform cross-section and is subjected to pure torque without significant bending or axial loading.

3. Can the same equations be used with a square shaft? Explain.
No, the standard torsion equations used in this experiment are derived for circular shafts. A square shaft experiences warping and a different shear stress distribution, so different torsion relationships must be used.

4. In our test, we used solid shafts. Could the same procedure be used with a hollow shaft? Explain.
Yes, the same general testing procedure can be used for a hollow circular shaft, but the polar moment of inertia must be changed to account for the inner and outer diameters:

$$J = \frac{\pi}{32}\left(D_o^4 - D_i^4\right)$$

Where D_o is the outer diameter and D_i is the inner diameter

5. How does shearing strain vary with the distance from the axis of the shaft?
Shear strain increases linearly with distance from the center axis of the shaft:

$$\gamma = \frac{r\theta}{L}$$

Therefore, the shear strain is zero at the center and reaches its maximum value at the outer surface.

6. Large shafts for ocean liners are often made hollow. Explain the reason for this.
Large shafts are made hollow because material near the center contributes relatively little to torsional strength. Removing this material reduces weight while maintaining a large portion of the shaft's torsional strength and stiffness. This makes the shaft lighter and more material-efficient.

Conclusions:

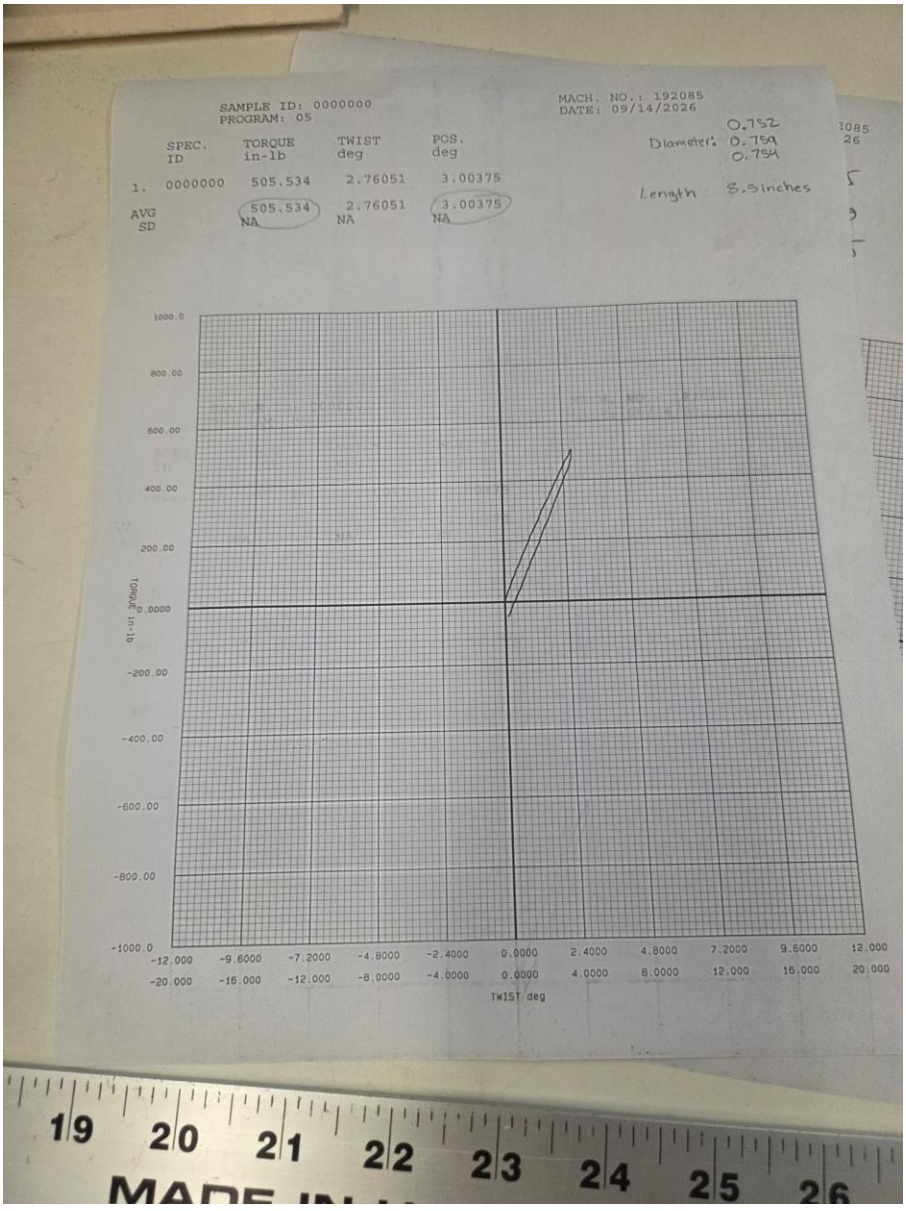
In this experiment, for steel and aluminum, the torsional behavior and shear modulus of elasticity (G) were evaluated. Circular rods of steel and aluminum were subjected to pure torque using a Lo-Torq torsion testing machine. The shear and shear strain were calculated for each material, within the linear-elastic range, allowing for the construction of their respective shear-strain diagrams. The experimental shear modulus of elasticity resulted in 45,566.47 psi for aluminum specimen and 133,626.36 psi for steel specimen. These results demonstrate a clear physical difference between the two metals: the steel rod demonstrating a significantly steeper stress-strain slope compared to aluminum, yielding a shear modulus roughly three times higher. The confirmation of steel

possessing a higher torsional stiffness and greater resistance to elastic shear deformation under applied torque, compared to aluminum is solidified. Thus, the fundamental theoretical assumption of circular shaft torsion was successfully illustrated. Possible sources of experimental error stem from minor measurement inaccuracies in initial specimen dimensions using the digital calipers and ruler, as well as calibration variances in angle and torque readouts from the testing equipment. In conclusion, the experiment effectively validated Hooke's law in torsion.

References:

1. https://www.engineeringtoolbox.com/young-modulus-d_417.html#gsc.tab=0

Appendices:

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2) L

SAMPLE ID: 0000000
PROGRAM: 05

MACH. NO.: 192085
DATE: 09/14/2026

	SPEC. ID	TORQUE in-lb	TWIST deg	POS. deg
1.	0000000	640.918	1.71337	2.01375
AVG		640.918	1.71337	2.01375
SD		NA	NA	NA

Diameter: 0.7485
0.7460
0.7535

length: 13in

